MILITARY SPECIFICATION

WARHEAD SECTION, ATOMIC, TRAINING: M72E1

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of training warhead section entitled Warhead Section, Atomic Training: M72E1, which is used to train personnel on the Honest John warhead section.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape, Pressure - sensitive, Adhesive, Waterproof, for Packaging and Packing.

MILITARY

MIL-S-48728(PA) - Selector, Fuze Option Assembly, Training.
MIL-T-48519(PA) - Timer, Sequential: M16 and M30.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8800664 - Control Assembly, Adaption Kit.
8800667 - Timer, Sequential, Training: M16.

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8813463 - Plug, Desiccant.
8821912 - Ring Support.
8821965 - Forward, H-L, Selector Assembly.
8822388 - Ballast, Cylinder, Control Assembly, Training.
8822392 - Selector, Fuze Option Assembly, Training.
8830817 - Shell and Frame Assembly, Middle Section.
8830823 - Warhead Section, Atomic Training: M72E1.
8830824 - Structure, Nose Base Assembly.
8830825 - Structure, Nose Tip Section.
8830832 - Timer, Sequential, Training: M30.
8833481 - Shipping and Storage Container, Atomic Warhead Section, M480E1.
8836379 - Ballast, Training.
EL-8867283 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The warhead section shall comply with all requirements specified on the Drawing (dwg) 8830823, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

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3.5 Continuity (pre-fire panel). The warhead section shall cause the pre-fire panel to give "Go" and "No-Go" indications as prescribed in Table IV when tested as specified in 4.4.1.

3.6 Operational tests.

3.6.1 Fuze option selector. The selector shall be capable of rotating freely in the clockwise (cw) and counterclockwise (ccw) directions, and shall permit no stopping between detent positions when testing as specified in 4.4.2.1.

3.6.2 Timer. The timer mechanism shall be capable of being set at any position from 0000 through 1299 to 0000 seconds, when testing as specified in 4.4.2.2.

3.6.3 Pressure retention. The valve and cylinder assembly shall be capable of retaining an air pressure of 15 + 2 pounds per square inch gage (psig) when testing as specified in 4.4.2.3.

3.6.4 Hob selector ("B" section). The height of burst (HOB) selector shall be capable of selecting "H" and "L" settings when testing as specified in 4.4.2.4.

3.7 Paint adhesion. Test for paint adhesion of the warhead section shall be as specified in 4.4.3. The item being tested shall exhibit no bare metal or underlying coating.

3.8 Weight and center of gravity. The warhead section shall weigh 1235 plus 15 pounds or minus 10 pounds, when fully assembled and the center of gravity shall be located at station 80 + 2 inches when tested as specified in 4.4.4.

3.9 Torque. The warhead section shall meet the assembling and mounting torque as prescribed in Table V when tested as specified in 4.4.5.

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460 (MU).

3.11 Timer M16 and M30 and selector fuze option assembly. The timer (dwg 8800667) and Selector (dwg 8822392) shall have complied with MIL-T-48519(PA) and MIL-S-48728(PA) before being assembled to the warhead section M72E1.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.12.2 Threads. Threads shall be full and undamaged for the entire minumum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8830823) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity (pre-fire panel) (see 3.5)		X
3. Operational tests (see 3.6)		X
4. Paint adhesion (see 3.7)		X
5. Weight and center of gravity (see 3.8)		X
6. Torque (see 3.9)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any if the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	37-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defoned in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Structure, nose base section (see dwg 8830824).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Finish improper	Visual
	102. Any component missing, damaged or loose	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.2 Shell and frame assembly, middle section (see dwg 8830817).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Finish improper	Visual
	102. Any component missing, damaged or loose	Visual/Manual
	103. Hardware missing, damaged or loose	Visual/Manual
	104. Access door bound	Manual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.3 Forward H-L selector assembly (see dwg 8821965).

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Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any component missing, damaged or loose	Visual/Manual
	102. Finish improper	Visual
	103. Hardware missing, damaged, or loose	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.4 Structure, nose tip section (see dwg 8830825).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Finish improper	Visual
	102. Any component missing, damaged, or loose	Visual/Manual
	103. Hardware missing, damaged, or loose	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.5 Ring support (see 8821912).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:		
	201. Incorrect outside diameter	Gage
	202. Incorrect thickness	Gage
	203. Scratch in protective finish with base metal exposed	Visual
	204. Marking illegible, incomplete or missing	Visual
	205. Color of mark other than white	Visual

4.3.2.1.6 Ballast training (see dwg 8836379).

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Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter at -D- not concentric with reference -C- and not perpendicular with reference -A-	Gage
102.	Rear diameter not concentric with reference -C- and not perpendicular with reference -A-	Gage
103.	Front surface -E- not parallel with reference -A-	Gage
104.	Incorrect diameter at -C-	Gage
105.	Incorrect rear inside diameter	Gage
106.	Incorrect front inside diameter	Gage
107.	Pitch diameter (max) (8) not perpendicular with reference -D-	Gage
108.	Incorrect depth of taped hole (8)	Gage
109.	Incorrect overall length	Gage
110.	Incorrect location of hole to rear surface	Gage
111.	Incorrect diameter of hole (16)	Gage
112.	Hole (16) not equally spaced	Gage
113.	Hole (16) not perpendicular with reference -A-	Gage
114.	Incorrect diameter of hole (3)	Gage
115.	Mislocation of hole to front surface	Gage
116.	Incorrect diameter of large hole (1) near front end	Gage
117.	Incorrect dimension to shoulder from rear surface	Gage
118.	Mislocation of hole (3) to front surface	Gage
119.	Mislocation of hole to front surface (12.80)	Gage
120.	Mislocation of hole to front surface (14.80)	Gage
121.	Two holes not located within prescribed tolerance	Gage
122.	Incorrect weight	*Gage

*Applicable only when ballast training is procured as a space part.

Minor:		
201.	Improper radius	Visual
202.	Marking illegible, improper or missing	Visual
203.	Color of mark other than white	Visual

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204. Scratch in protective finish with base metal exposed.....Visual
205. Paint other than black.....Visual
206. Evidence of poor workmanship (see 3.12).....Visual
207. Improper finish.....Visual

4.3.2.1.7 Warhead section, atomic training: M72E1 (see dwg 8830823).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any operation missing or inadequate	Visual
102.	Soldering improper (see 3.10)	Visual
103.	Marking missing, illegible, or improper	Visual
104.	Nose tip bolt missing, damaged, or loose	Visual/Manual
105.	Any component missing, damaged, or loose	Visual/Manual
106.	Objective evidence relative to material missing or improper (see 3.1)	Visual
107.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.8 Preparation for delivery (see dwg 8833481).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing in accordance with dwg 8833481	Visual
102.	Marking in accordance with dwg 8833481	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.2)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

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Test	Test classification
a. Continuity (pre-fire panel)	Major
b. Operational tests	Major
c. Paint adhesion	Major
d. Weight and center of gravity	Major
c. Torque	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity (pre-fire panel), operational tests, paint adhesion, weight and center of gravity, and torque. Each warhead section shall be tested in accordance with 4.4.1 through 4.4.5. If any item fails to comply with 3.5 through 3.9, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8867283). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerance used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity (pre-fire panel). The prefire test socket shall be manually rotated, at a normal rate, to the positions indicated in Table IV, by using a Prefire Adjustable Tool X4137.

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Table IV

Test No.	Warhead Section Switch Position	Generation Rotation Direction use Pre-fire test Socket	Indication
1	No-Go	Counterclockwise (ccw)	Both lamps light
2	Go	Counterclockwise	Both lamps light
3	No-Go	Clockwise (cw)	Only red lamp lights
4	Go	Clockwise	Only green lamp lights

4.4.2 Operational tests.

4.4.2.1 Fuze option selector. Insert wrench H-4137 into the receiving slot and rotate the T-Handle cw and ccw. slowly, several times. During this rotating operation observe that the selector does not assume a "non-indicating" position, i.e. - a position other than that indicating "GO" or "NO-GO".

4.4.2.2 Timer. Insert wrench H-4137 into the receiving slot and rotate the T-Handle until the dial completes the full cycle from 0000 through 1299 to 0000.

4.4.2.3 Pressure rotation. Fill the valve and cylinder assembly with dry air to 15 ± 2 psig. After 5 minutes have elapsed, check the air pressure with the Government approved Test Set T-283 to determine loss of air pressure. NOTE: If a waiver of the T-283 is required refer to 6.3.

4.4.2.4 Hob selector ("B" section). Insert wrench H-4137 into the receiving slot of the selector and rotate the T-Handle in the cw and ccw directions several times. During this rotating operation observe that the selector does not assume any position other than "L" or "H".

4.4.3 Paint adhesion. This test shall be performed on all sections of the warhead (see dwg 8830823) after a minimum of 24 hours drying unless otherwise specified. Water-resistance, pressure sensitive adhesive tape (3/4 inch width) conforming to the requirments of Specification PPP-T-60, Type II or III, Grade A or B shall be used. Press a 2 inch length of a somewhat longer piece firmly onto a flat or cylindrical surface of the item rubbing out all air bubbles under the tape. Allow approximately 10 seconds for the test area to return at room temperature. Grasp a free end of the tape and at a rapid speed strip it from the item or specimen by pulling the tape back upon itself 180 degrees in such a manner that the tape is folded back to back during the procedure. Observe for bare spots where the paint is removed.

4.4.4 Weight and center of gravity. The weight shall be determined by using a scale capable of weighing within 5 pounds.

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The center of gravity will be determined by the moment method using equipment that has been approved by the contracting officer see 6.3.

4.4.5 Torque. Screws and bolts, used to mount and assemble the items listed in Table V shall be fastened down with the moments prescribed therein. The torque shall be measured with a torque wrench suitably adapted to the hardware being assembled.

Table V

Item	Item to Which Mated	Torque Required (inch-pounds)
Ballast training (8836379)	Control assembly, adaption kit (8800664)	120 ± 15
Timer, sequential, training: M16 (8800667) or Timer, sequential, training: M30 (8822388)	Control assembly, adaption kit (8800664)	50 ± 5
Ballast, cylinder, control assembly training (8822388)	Structure, nose, base, section (88308824)	175 ± 18
Structure, center section (8830817)	Structure, nose, base, section (88308824)	175 ± 10
Forward, H-L selector assembly (8821965)	Structure, center, section (8830817)	115 ± 10**
Desiccant plug (8813463)	Control assembly, adaption kit (8800664)	20 to 25

**The contractor shall certify over his signature, that the torque requirement has been complied with.

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5 PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The warhead section shall be packaged in accordance with dwg 8833481 using the specified moisture barrier.

5.1.2 Level C. The warhead section shall be packaged in accordance with dwg 8833481 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The warhead section shall be packed in accordanc with dwg 8833481.

5.3 Marking. Marking shall be as specified in accordance with dwg 8833481.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated on the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1816 (Rev 2) Dated 1 April 1964 With Amendment 2 Dated August 18 1965

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A051

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